Day 004 | Unit 01 | Teacher plan

Positive Rational Numbers: Decimals and Division

Divide Whole Numbers

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 12

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01 / Learn

Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

Worked example 1

Use long division: 304 ÷ 4. Show the algorithm and check.

1. Estimate how many digits the quotient should have. Work from the leftmost digits of the dividend.

2. 30 ÷ 4: write 7, subtract 28, remainder 2

3. bring down the next digit: 24 ÷ 4: write 6, subtract 24, remainder 0.

4. Check: 4 × 76 = 304.

Answer: 76

Worked example 2

Use long division: 2,948 ÷ 23. Show the algorithm and check.

1. Estimate the quotient first.

2. 29 ÷ 23: write 1, subtract 23, remainder 6

3. bring down the next digit: 64 ÷ 23: write 2, subtract 46, remainder 18

4. bring down the next digit: 188 ÷ 23: write 8, subtract 184, remainder 4.

5. Check: 23 × 128 + 4 = 2948.

Answer: 128 R 4

Day 004 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use long division: 536 ÷ 8. Show the algorithm and check.

02 / Constructed response

Use long division: 256 ÷ 4. Show the algorithm and check.

03 / Constructed response

Use long division: 592 ÷ 8. Show the algorithm and check.

04 / Constructed response

Use long division: 6,194 ÷ 19. Show the algorithm and check.

05 / Constructed response

Use long division: 7,854 ÷ 22. Show the algorithm and check.

06 / Constructed response

Use long division: 4,008 ÷ 12. Show the algorithm and check.

07 / Constructed response

Use long division: 4,255 ÷ 17. Show the algorithm and check.

08 / Model and constructed response

Use long division: 15,080 ÷ 37. Show the algorithm and check.

09 / Model and constructed response

Use long division: 2,900 ÷ 19. Show the algorithm and check.

10 / Constructed response

A club has 5132 cards. Each full packet holds 13. How many full packets can it make, and how many cards remain?

Day 004 | Unit 01 | Answer key

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1. 67

53 ÷ 8: write 6, subtract 48, remainder 5; bring down the next digit: 56 ÷ 8: write 7, subtract 56, remainder 0. Check: 8 × 67 = 536.

6.NS.B.2 | Constructed response

2. 64

25 ÷ 4: write 6, subtract 24, remainder 1; bring down the next digit: 16 ÷ 4: write 4, subtract 16, remainder 0. Check: 4 × 64 = 256.

6.NS.B.2 | Constructed response

3. 74

59 ÷ 8: write 7, subtract 56, remainder 3; bring down the next digit: 32 ÷ 8: write 4, subtract 32, remainder 0. Check: 8 × 74 = 592.

6.NS.B.2 | Constructed response

4. 326

61 ÷ 19: write 3, subtract 57, remainder 4; bring down the next digit: 49 ÷ 19: write 2, subtract 38, remainder 11; bring down the next digit: 114 ÷ 19: write 6, subtract 114, remainder 0. Check: 19 × 326 = 6194.

6.NS.B.2 | Constructed response

5. 357

78 ÷ 22: write 3, subtract 66, remainder 12; bring down the next digit: 125 ÷ 22: write 5, subtract 110, remainder 15; bring down the next digit: 154 ÷ 22: write 7, subtract 154, remainder 0. Check: 22 × 357 = 7854.

6.NS.B.2 | Constructed response

6. 334

40 ÷ 12: write 3, subtract 36, remainder 4; bring down the next digit: 40 ÷ 12: write 3, subtract 36, remainder 4; bring down the next digit: 48 ÷ 12: write 4, subtract 48, remainder 0. Check: 12 × 334 = 4008.

6.NS.B.2 | Constructed response

7. 250 R 5

42 ÷ 17: write 2, subtract 34, remainder 8; bring down the next digit: 85 ÷ 17: write 5, subtract 85, remainder 0; bring down the next digit: 5 ÷ 17: write 0, subtract 0, remainder 5. Check: 17 × 250 + 5 = 4255.

6.NS.B.2 | Constructed response

8. 407 R 21

150 ÷ 37: write 4, subtract 148, remainder 2; bring down the next digit: 28 ÷ 37: write 0, subtract 0, remainder 28; bring down the next digit: 280 ÷ 37: write 7, subtract 259, remainder 21. Check: 37 × 407 + 21 = 15080.

6.NS.B.2 | Model and constructed response

9. 152 R 12

29 ÷ 19: write 1, subtract 19, remainder 10; bring down the next digit: 100 ÷ 19: write 5, subtract 95, remainder 5; bring down the next digit: 50 ÷ 19: write 2, subtract 38, remainder 12. Check: 19 × 152 + 12 = 2900.

6.NS.B.2 | Model and constructed response

10. 394 full packets; 10 cards remain

5132 = 13 × 394 + 10.

6.NS.B.2 | Constructed response